

Emmanuel Cobian

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science

Berkeley, CA

GPA: 3.8

EXPERIENCE

Software Engineering Intern

Jan 2026 – Jun 2026

EquityBound

Berkeley, CA

- Diagnosed and resolved a MongoDB geospatial performance bottleneck on the business map endpoint by adding a `2dsphere` index, switching to `$geoWithin` queries, cleaning corrupted geo data, adding Redis caching, and parallelizing backend fetches, cutting query execution time from 60ms to 13ms (4.6x faster) and reducing documents scanned by 95.7%.
- Shipped a new county demographics feature on the interactive map, adding hover tooltips that surface sourced figures (median household income, crime rate, median age) for any selected county.
- Drove a cross-browser, cross-device UAT bug-fixing effort spanning 6+ major product features (Monthly Books, Cash Intelligence, Debt Management, Client Dashboard, Valuation), resolving high- and medium-severity layout, overflow, and contrast issues on Chrome, Safari, and Firefox across mobile, tablet, and desktop.

Software Engineering Intern

May 2025 – Aug 2025

Goplai

Berkeley, CA

- Fine-tuned custom YOLOv11 and SlowFast models for basketball player, ball, and hoop detection, manually annotating 500+ frames and clips to create high-quality datasets.
- Built a cloud-native, real-time basketball analysis API with FastAPI, integrating detection, player tracking, scoring, and ball possession logic, with AWS S3 storage and WebSocket streaming.
- Deployed application in Docker containers on AWS EC2 with automated health checks and systemd service orchestration, ensuring scalable, production-grade reliability.

Software Developer

May 2023 – May 2024

UC Berkeley MBA Program for Executives

Berkeley, CA

- Developed a Next.js web application to streamline the creation of personalized teams, reducing manual task completion time from days to minutes, achieving a 400x efficiency improvement.
- Designed an approximation algorithm to dynamically generate optimally balanced teams based on team size, gender diversity, experience levels, and work history.

Tech Fellow

Sep 2021 – May 2023

UC Berkeley Career Center

Berkeley, CA

- Tested and implemented a Python script using BeautifulSoup and Selenium to scrape content from over 1000+ employer websites to search for DEI Employment keywords.
- Parsed, analyzed, and visualized 200k+ data points using the pandas library in Python to provide tangible and understandable solutions to a non-technical audience.

PROJECTS

ICE Breakers | *React Native, Tailwind CSS, Firebase Auth, Express.js, Git*

- Developed a React Native mobile app that provides multilingual emergency support for immigrant communities, featuring a panic button, Red Card tool, and local resource integration.

Rate My Classes | *React, NextAuth.js, Python, MySQL, Selenium, BeautifulSoup, Git*

- Built a full-stack web app enabling UC Berkeley students to rate/review courses. Implemented secure authentication with NextAuth.js and a relational MySQL Database.

Exposify | *Spotify API, React, Next.js, Git*

- Music insights web app integrating the Spotify API. Built a backend with Next.js API routes to analyze top artists/playlists and generate personalized profiles.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C/C++, SQL, HTML/CSS, Go

Frameworks: React, Next.js, Node.js, Bootstrap, express.js

Developer Tools: AWS (EC2, S3), Docker, Postman, Git, Google Cloud Platform, VS Code, MongoDB, Redis

Libraries: pandas, NumPy, Matplotlib, SciPy, Scikit-learn, OpenMP, BeautifulSoup, Selenium